

MODELING OF HEAT TRANSFER AND FLUID FLOW DURING LASER WELDING OF ALUMINUM

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Abstract — This paper presents a two-dimensional model to study the fluid flow and heat transfer phenomena which occur during deep penetration laser welding. Using finite volume method analysis software FLUENT, behavior of temperature field and weld pool shape were analyzed. The results show that the method can be used to predict the shape of the welding pool as function of welding parameters and that the widths of both the fusion zone and heat effected zone decrease as the welding speed increases.

Keywords: laser welding, phase change, keyhole, FLUENT

INTRODUCTION

In deep penetration laser welding, when a laser beam with high intensity irradiates, a keyhole is formed in the workpiece, which enables the laser beam to penetrate deeply into the workpiece. As shown in Figure 1, the laser beam is pointed perpendicularly to the workpiece, the latter is then moved relative to the laser at a constant speed. Dowden and *al* [1,2] proposed several two-dimensional, analytical and numerical, models of temperature distribution in the horizontal plane of laser welding with fluid flow. Most of their models assumed that the heat transfer and fluid flow are two dimensional and the keyhole is circular and isothermal. Mahrle and *al* [3] used a two dimensional approach to solve the continuity and momentum equations to obtain the melt flow field around the vapor hole as a substitution of the assumed potential flow velocity distribution. The solid- liquid phase change was handled by enhancing the values of melt specific heat near the melting temperature. This paper presents a two-dimensional analysis of heat transfer and fluid flow during deep laser welding of 7075 aluminum. The governing equations were solved with the computational fluid dynamics package FLUENT. The evolution of the phase change front is undertaken through an enthalpy formulation of the heat transfer problem. These models are mainly based on the fixed grid enthalpy based method [4].

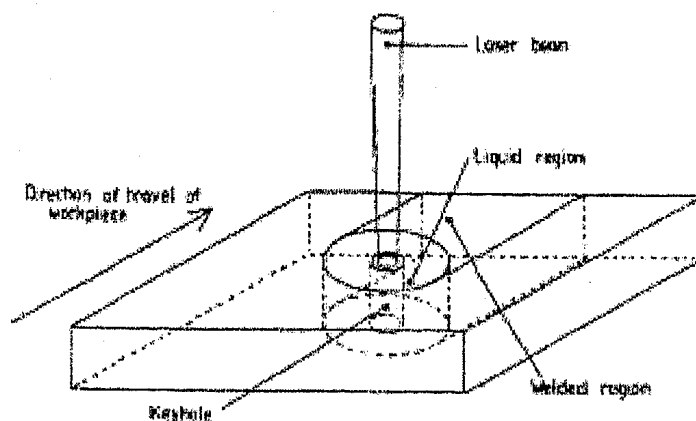


Fig 1 — Geometry of the laser beam relative to the workpiece.

MATHEMATICAL MODEL

In penetration welding with a laser beam a long, thin and almost cylindrical vapor cavity is reported to be formed through the workpiece, which is surrounded by a region of molten pool, most of which flows round the keyhole and freezes to rear to form a weld. We study two dimensional horizontal model as shown in figure 2 and we have the following assumptions:

- The heat transfer and fluid flow is two dimensional.
- The vapor cavity is circular and at vaporization temperature.
- The flow is laminar and incompressible.

Based on the foregoing assumptions, the governing equations describing the melt flow and heat transfer in the workpiece are as follows.

Continuity equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

momentum equations in x and y directions:

$$\rho \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + S_u \quad (2)$$

$$\rho \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + S_v \quad (3)$$

energy equation:

$$\rho C_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + S_h \quad (4)$$

where u and v are the velocity components in x and y direction, p and T are pressure and temperature, whereas ρ, μ, k and C_p are density, viscosity, thermal conductivity and specific heat, respectively. In order to account for the solid- liquid phase change in the weld pool, the fixed grid method suggested by Voller et al [4] is employed. In this method, interface are incorporated into the governing equations via introducing pertinent source terms. The source terms S_u, S_v introduced in the momentum equations (1) and (2) take the following forms:

$$S_u = A(u - U_s), \quad S_v = Av \quad (5)$$

Where A is function written as:

$$A = -\frac{C(1-f)^2}{(f^3 + q)} \quad (6)$$

Where $q=0.001$ is introduced to avoid any division by zero and C is constant and set equal to $1.64 \times 10^3 \text{ (kg / m}^3\text{s)}$, as suggested in [4].

S_h in equation (4) is introduced for the latent heat evolution at the solid liquid interface and takes the following form [4]:

$$S_h = -\rho L \quad (7)$$

Where L is the melting latent heat.

Boundary conditions employed in this study are as follow:

- At the keyhole boundary $r=a$,

$$v_r = 0, \quad \partial v_\theta / \partial r = 0, \quad T = T_v \quad (8)$$

- In the solid region:

$$u = U_s, \quad v = 0, \quad T = T_0 \quad (9)$$

where T_0 is the ambient temperature.

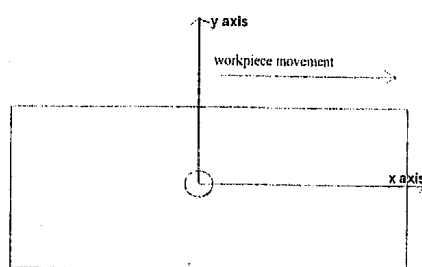


Fig 2 – The model for computation of the flow field

The basic equations were numerically solved using FLUENT Software (version 6.1). In the latter a finite volume method is developed to solve the Navier-Stokes and energy equations. For the discretisation of the convective term the QUICK difference scheme is used. The pressure velocity coupling is ensured by the SIMPLE algorithm.

RESULTS AND DISCUSSION

The material used in the present investigation is a type of 7075 Aluminum and the keyhole radius is taken to be 0.1 mm. Figure 3 show the temperature distribution for three different welding speed. Results show that the isotherms in the front of the keyhole are closer than those behind the keyhole because of the convection effect of the welding speed movement. It is also important to observe that the width of the heat affected zone in the workpiece is reduced as the welding speed increases. Figure 4 depicts the sizes of the weld pool as function of welding speed. Since the keyhole moves, the weld pool shape is not circular as in the case of a stationary heat source. The elliptic nature of the pool shape is more prominent for the higher welding speed case. It can be seen also, that the weld pool at a higher speed is smaller. This implies that a narrower fusion zone may be obtained as the welding speed increases. The energy absorbed by the keyhole in term of welding speed is shown in figure 5. The power intensity absorbed by the keyhole increases with increasing welding speed. Namely, more laser energy is required for the full penetration welding of the workpiece as the welding speed increases.

CONCLUSION

In this paper, a two dimensional model has been introduced to describe solid/liquid phase change transformation with a mushy region in laser welding process of type 7075 Aluminum. Coupling calculation of thermal and fluid field were performed with finite volume method software FLUENT. A parametric study indicates the importance of the welding speed on the widths of both the fusion zone and heat affected zone. The prediction relationship between the laser absorbed across the keyhole and welding speed can be used to evaluate the energy efficiency in laser full penetration welding.

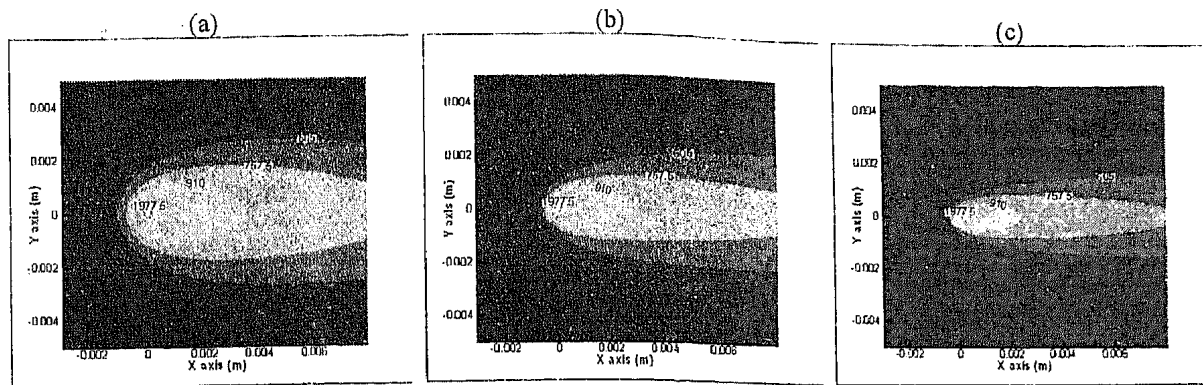


Fig 3 – Temperature distribution for three different welding speed.
(a) $U_s=0.05$ m/s, (b) $U_s=0.1$ m/s, (c) $U_s=0.2$ m/s

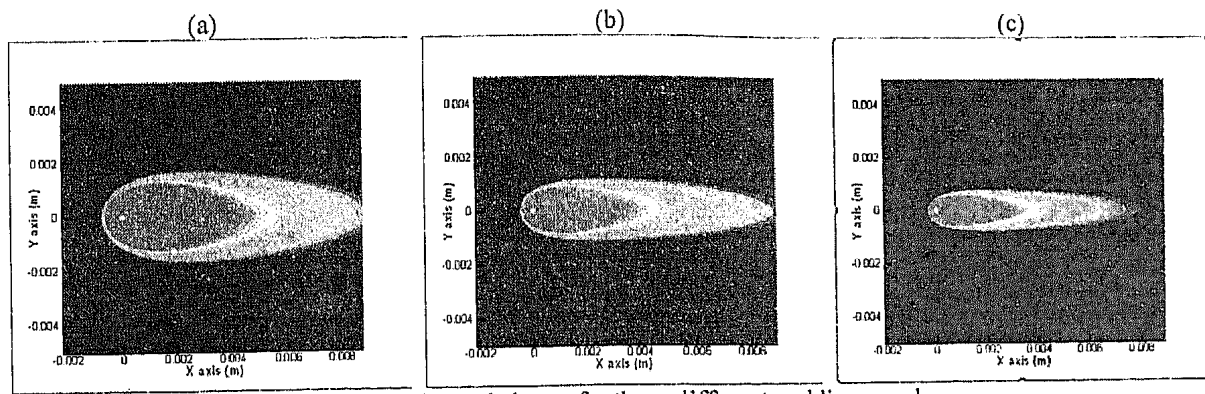


Fig 4 – Weld pool shapes for three different welding speed.
(a) $U_s=0.05$ m/s, (b) $U_s=0.1$ m/s, (c) $U_s=0.2$ m/s

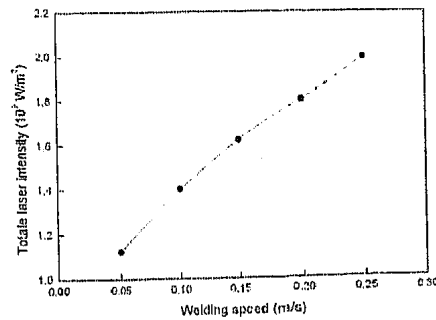


Fig 5 – Variation of totale laser intensity with the welding speed.

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